

SECSR2043 OPERATING SYSTEMS
[20 Marks]

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 Section : 3

Marks

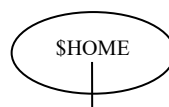
Instruction: Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*. *.; cp ../fruits/*. *.;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

```

GNU nano 7.2                               joanne.sh *
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .

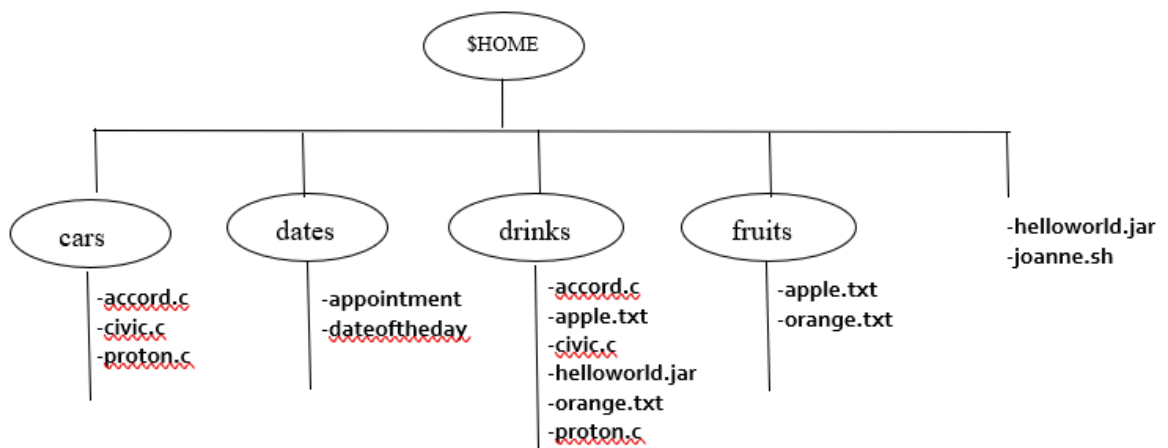
```

Then draw the tree

```

joanne@secre2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   '-- proton.c
|-- dates
|   |-- appointment
|   '-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   '-- proton.c
|-- echo.c
|-- fruits
|   |-- apple.txt
|   '-- orange.txt
|-- helloworld.jar
-- joanne.sh

```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 ifiles.sh *
#!/bin/bash

read -p "Enter the file extension (without dot):" ext

echo "Listing all *.$ext files in current directory and subdirectories:"
find . -type f -name "*.$ext"
count=$(find . -type f -name "*.$ext" | wc -l)
echo "Total number of *.$ext files: $count"

joanne@secr2043:~$ chmod u+x ifiles.sh
joanne@secr2043:~$ ./ifiles.sh
Enter the file extension (without dot):c
Listing all *.c files in current directory and subdirectories:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Total number of .c files: 6
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

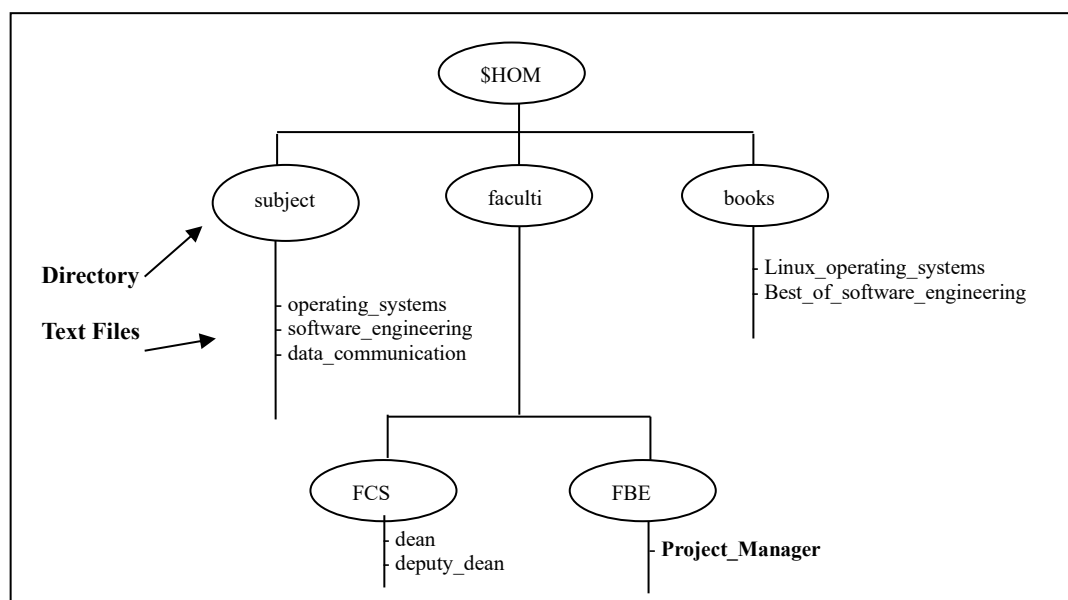


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2                               joanne2a.sh
mkdir -p $HOME/subject $HOME/faculti/FCS $HOME/faculti/FBE $HOME/books
echo "operating systems" > $HOME/subject/operating_systems
echo "software engineering" > $HOME/subject/software_engineering
echo "data communication" > $HOME/subject/data_communication

echo "dean" > $HOME/faculti/FCS/dean
echo "deputy dean" > $HOME/faculti/FCS/deputy_dean

echo "Project Manager" > $HOME/faculti/FBE/Project_Manager

echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering

joanne@secr2043:~$ chmod u+x joanne2a.sh
joanne@secr2043:~$ ./joanne2a.sh
joanne@secr2043:~$ tree $HOME
/home/joanne
|-- books
|   |-- Best_of_software_engineering
|   '-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   '-- proton.c
|-- dates
|   |-- appointment
|   '-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   '-- proton.c
|-- faculti
|   |-- FBE
|   |   '-- Project_Manager
|   '-- FCS
|       |-- dean
|       '-- deputy_dean
|-- fruits
|   |-- apple.txt
|   '-- orange.txt
|-- helloworld.jar
|-- ifiles.sh
|-- joanne.sh
|-- joanne2a.sh
|-- subject
|   |-- data_communication
|   |-- operating_systems
|   '-- software_engineering
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--



- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```

GNU nano 7.2                               joanne2c.sh *
#!/bin/bash

# Set access control for 'subjects' directory
chmod 754 $HOME/subject

# Set access control for 'Best_of_software_engineering' file
chmod 640 $HOME/books/Best_of_software_engineering

# Set access control for 'FCS' directory
chmod 607 $HOME/faculti/FCS

# Set access control for 'Project_Manager' file
chmod 026 $HOME/faculti/FBE/Project_Manager

joanne@secre2043:~/books$ ls -ld $HOME/subject
ls -l $HOME/books/Best_of_software_engineering
ls -ld $HOME/faculti/FCS
ls -l $HOME/faculti/FBE/Project_Manager
drwxr-xr-x 2 joanne joanne 4096 Jun 13 12:14 /home/joanne/subject
-rw-r--r-- 1 joanne joanne 29 Jun 13 12:32 /home/joanne/books/Best_of_software_engine
drw----rwx 2 joanne joanne 4096 Jun 13 12:14 /home/joanne/faculti/FCS
-----wx-x 1 joanne joanne 16 Jun 13 12:32 /home/joanne/faculti/FBE/Project_Manager
  
```



file after executing the bash script in question 2(c)).

[2 marks]

Directory/File	Access Control
subjects	drwxr-xr--
Best_of_software_engineering	-rw-r-----
FCS	drw----rwx
project_manager	--w--wx--x

End of Lab 3

**** All the Best for Final Exam ****